

exp 5 emff.sci (C:\Users\pachl\Documents\exp 5 emff.sci) - SciNotes

Untitled 1 exp 5 emff.sci

```
28
29 disp("The value of V2 =");
30 disp(v2);
31
32 // Potential difference
33 v = v1 - v2;
34
35 disp("The electric potential difference =");
36 disp(v);
37
38 // Graph
39 r1g = linspace(0,10,50);
40 r2g = linspace(0,10,50);
41 rg = linspace(0,10,50);
42
43 y = v1*exp(-0.1*r1g);
44 z = v2*exp(-0.1*r2g);
45 x = v*exp(-0.1*rg);
46
47 figure();
48
49 subplot(2,2,1);
50 plot2d3(r1g,y);
51 xlabel("Radius-1");
52 ylabel("Voltage V1");
53 title("Potential V1");
54
55 subplot(2,2,2);
56 plot2d3(r2g,z);
57 xlabel("Radius-2");
58 ylabel("Voltage V2");
59 title("Potential V2");
60
61 subplot(2,2,3);
62 plot2d3(rg,x);
63 xlabel("Radius");
64 ylabel("Voltage Difference");
65 title("Potential Difference");
```

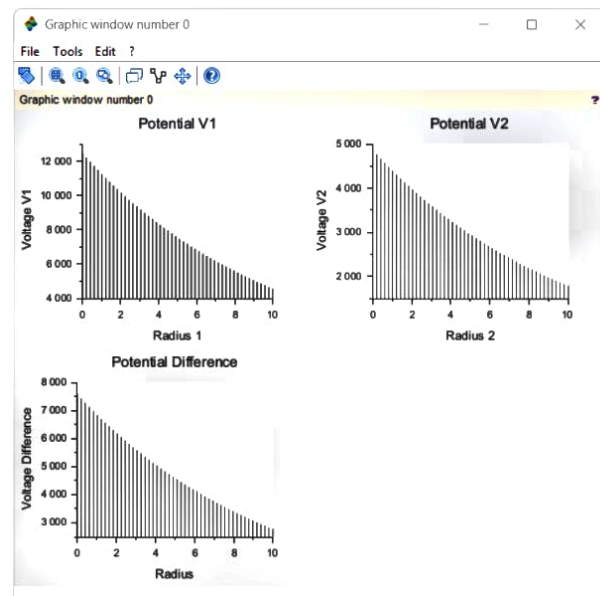
SciLab 2025.1.0 Console

File Edit Control Applications ?

SciLab 2025.1.0 Console

```
"The value of V1 ="
12463.756
"The value of V2 ="
4874.2877
"The electric potential difference ="
7589.4685
```

--> |



exp 5:-

$$q = 5 \times 10^{-6}$$

$$a_x = 2$$

$$a_y = 3$$

$$b_x = 6$$

$$b_y = 7$$

$$r_1 = \sqrt{2^2 + 3^2}$$

$$r_1 = 3.606$$

$$r_2 = \sqrt{6^2 + 7^2}$$

$$r_2 = 9.220$$

i) Find V_1

$$V_1 = \frac{q}{4\pi\epsilon_0 r_1}$$

$$V_1 = \frac{5 \times 10^{-6}}{4 \times 3.1416 \times 8.854 \times 10^{-12} \times 3.606}$$

$$V_1 = 1249.5 \text{ V}$$

ii) Find V_2

$$V_2 = \frac{q}{4\pi\epsilon_0 r_2}$$

$$V_2 = \frac{5 \times 10^{-6}}{4 \times 3.1416 \times 8.854 \times 10^{-12} \times 9.220}$$

$$V_2 = 4888.7 \text{ V}$$

iii) potential

$$V = V_1 - V_2$$

$$V = 1249.5 - 4888.7$$

$$V = -3639.2 \text{ V}$$